Goodreads Book Data Analysis

Project 3

Prepared by: Fatemeh Shahabdehkordi

ALY 6000

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Introduction

This report presents an analysis of book data collected from Goodreads, a popular book review and recommendation website, originally archived on Kaggle. The dataset originally contained 52,448 books with 23 variables. After thorough data cleaning including filtering to books published between 1990 and 2020, removing books exceeding 700 pages, standardizing column names, converting date formats, and removing rows with missing values the final dataset consisted of 8,490 books across 17 variables.

The primary objective of this project was to explore patterns in book ratings, page counts, publication trends over time, and publisher dominance using descriptive statistics and data visualizations. All analysis was conducted in R using the tidyverse, ggplot2, janitor, and lubridate packages.

Key Statistical Findings

Measures of Central Tendency and Dispersion

Descriptive statistics were computed for the key numerical variables in the dataset. The mean is the arithmetic average of all values, calculated by summing all observations and dividing by the total count. The median is the middle value when all observations are sorted in order and is less sensitive to extreme values (outliers) than the mean. Together, the mean and median describe where most of the data is centered.

Dispersion refers to how spread out the data values are. The minimum and maximum values define the full range of the data, while the interquartile range (IQR) — the difference between the 1st quartile (Q1) and 3rd quartile (Q3) — captures the spread of the middle 50% of observations. The key statistics from the cleaned dataset are summarized below:

```
Book Ratings:
  Minimum: 1.99      Maximum: 5.00
  Mean: 4.02         Median: 4.03
  Q1: 3.86           Q3: 4.19

Page Counts:
  Minimum: 0         Maximum: 699
  Mean: 329.7        Median: 328
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Q1: 240          Q3: 405

Liked Percent:   Mean: 92.87%     Median: 94%
Publication Year:   Range: 1990-2020     Median Year: 2007
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Book ratings showed a mean of 4.02 and a median of 4.03. Since these two values are nearly identical, this indicates a roughly symmetric distribution , most books cluster around 4.0 with very few receiving extremely low ratings. The average liked percentage of 92.87% further confirms that Goodreads readers tend to engage positively with the books they review. Page counts had a mean of 329.7 and a median of 328, again very close, confirming a symmetric distribution of book lengths.

Visualization Analysis

Figure 1: Histogram of Book Ratings

A histogram displays the frequency distribution of a continuous variable by dividing data into equal-width bins. In this analysis, a bin width of 0.25 was used, meaning each bar represents a 0.25-point rating range.

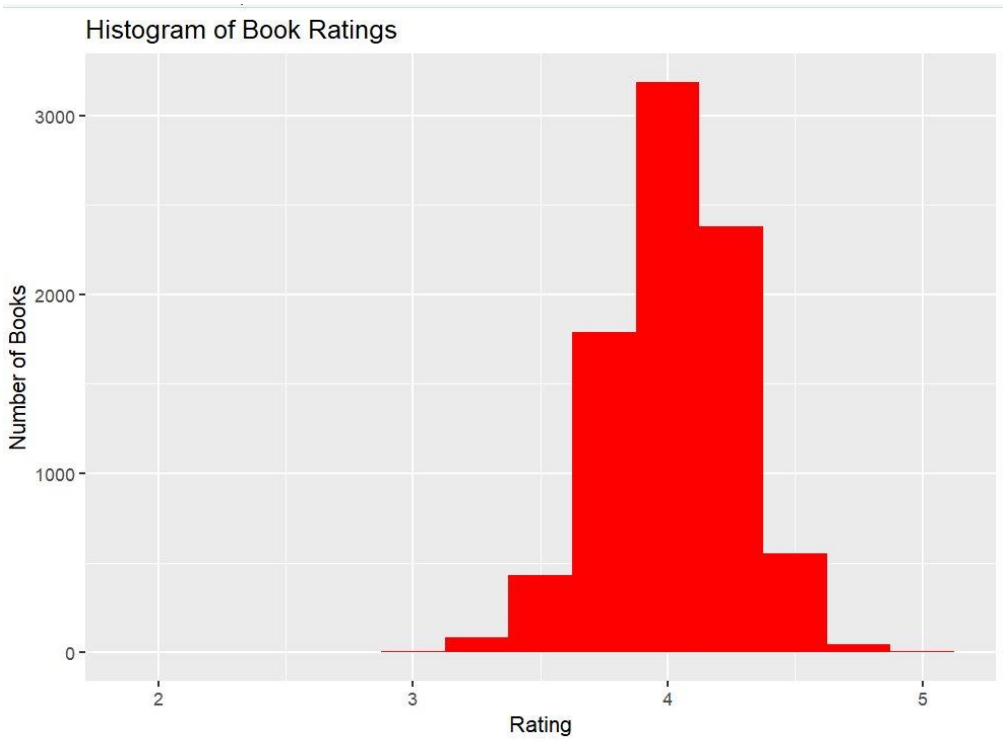

Figure 1. Histogram of Book Ratings (bin width = 0.25)

As shown in Figure 1, the vast majority of books received ratings between 3.75 and 4.50, with the tallest bar in the 4.00-4.25 bin containing approximately 3,200 books. The distribution is unimodal (one clear peak) and slightly left-skewed, with the tail extending toward lower ratings. Very few books received ratings below 3.0 or above 4.75. This pattern is consistent with the mean of 4.02 and suggests Goodreads users are generally generous raters, likely because readers self-select books they expect to enjoy before reviewing them.

Figure 2: Box Plot of Page Counts

A boxplot summarizes a variable using five key statistics: minimum, Q1, median, Q3, and maximum. The box spans from Q1 to Q3 (IQR), the line inside marks the median, whiskers extend to 1.5 times the IQR, and values beyond the whiskers appear as individual outlier points.

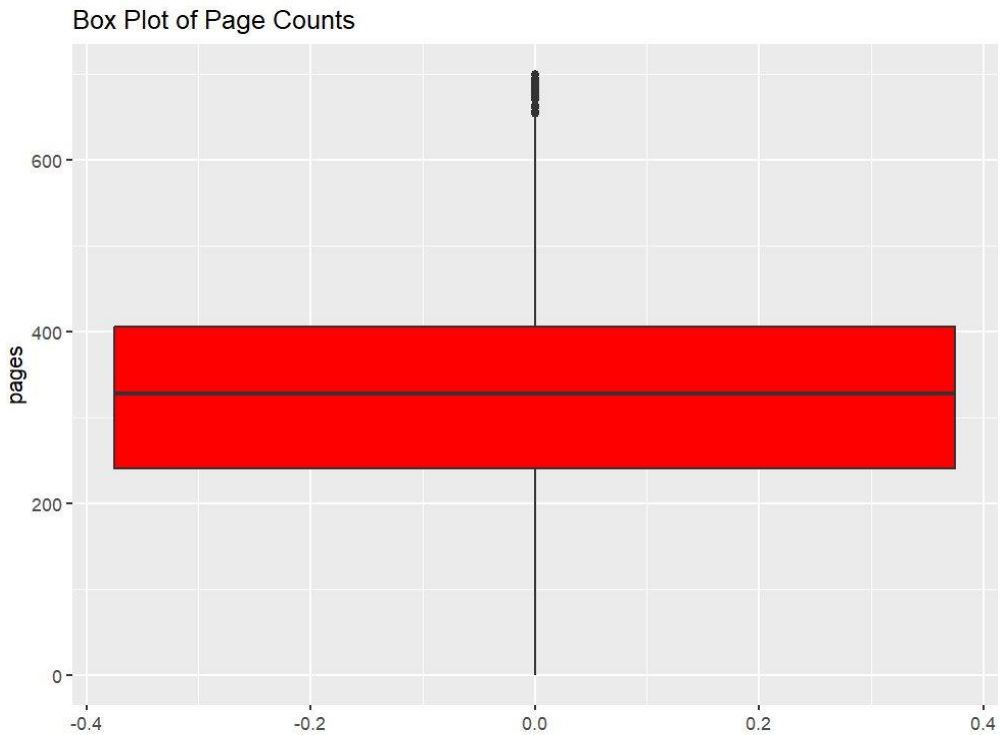

Figure 2. Horizontal Box Plot of Page Counts

As shown in Figure 2, the median page count is approximately 328 pages, with the box spanning from 240 (Q1) to 405 (Q3) pages, an IQR of 165 pages. The left whisker extends toward 0 pages while the right whisker reaches approximately 650 pages. A cluster of outlier points near 650-699 pages represents unusually long books approaching the 700-page filter limit. The symmetric box shape confirms the balanced distribution noted in the summary statistics.

Figure 3: Total Number of Books Rated Per Year (1990-2020)

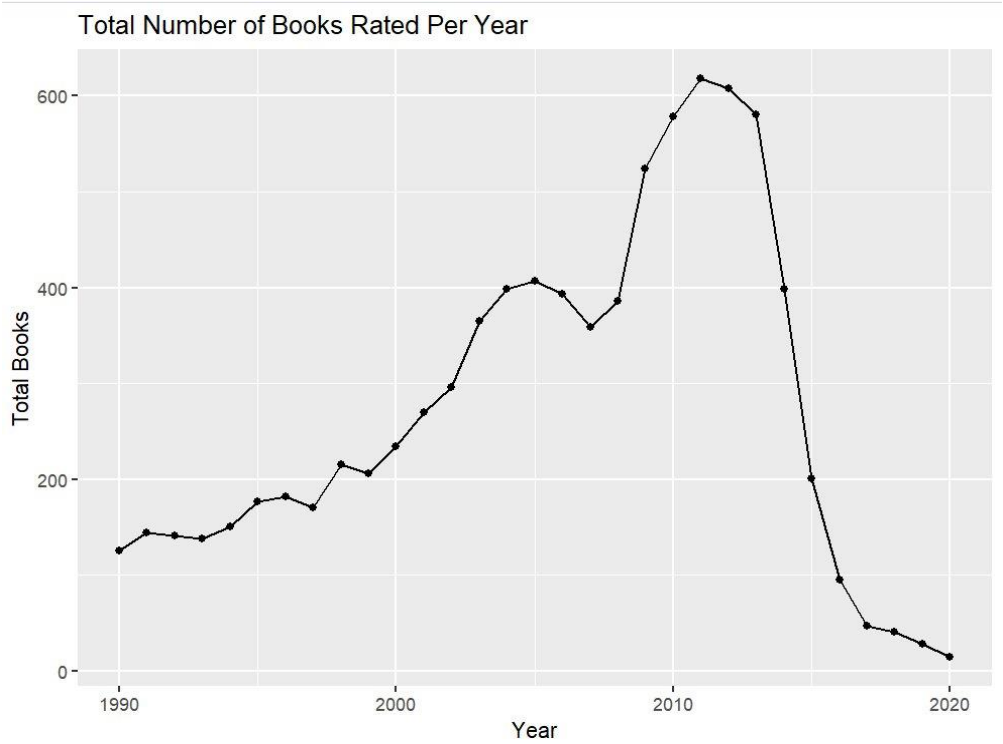

Figure 3. Total Number of Books Rated Per Year (1990-2020)

A line plot with points displays trends over time. Each point represents the total number of books first published in that year, and the connecting line makes it easy to follow the overall trend across the full 30-year period.

As shown in Figure 3, there is a clear growth trend from 1990 through 2011, rising from around 125 books in 1990 to a peak of approximately 625 books in 2011. This reflects both the growth of the publishing industry and the rise of Goodreads as a platform. Following the 2011 peak, a sharp decline is visible, with totals dropping to fewer than 25 books by 2020. This is a data artifact books published more recently have had less time to accumulate ratings and therefore appear less frequently in the dataset.

Figure 4: Pareto Chart — Book Counts by Publisher (1990-2020)

A Pareto chart combines a bar chart, and a cumulative line called an ogive to highlight which categories account for the largest share of a total. Bars are arranged in descending order from left to right, and the cumulative line shows the running total. A new data frame called book_publisher was created by grouping books by publisher and counting how many books each publisher contributed. Publishers with fewer than 120 books were removed to focus on major publishers only. Three calculated columns were then added: a cumulative count (cum_counts), a relative frequency (rel_freq) calculated by dividing each publisher's count by the total, and a cumulative frequency (cum_freq) as the running sum of relative frequencies. As shown in Figure 4, Random House is the dominant publisher with approximately 1,150 books, followed by Harper Collins with approximately 750 books. MacMillan, Hatchette, Simon and Schuster, and Scholastic Books follow with considerably fewer titles. The cumulative ogive rises steeply after the first two publishers and begins to flatten, confirming that Random House and Harper Collins together account for the largest share illustrating significant concentration in the publishing industry.

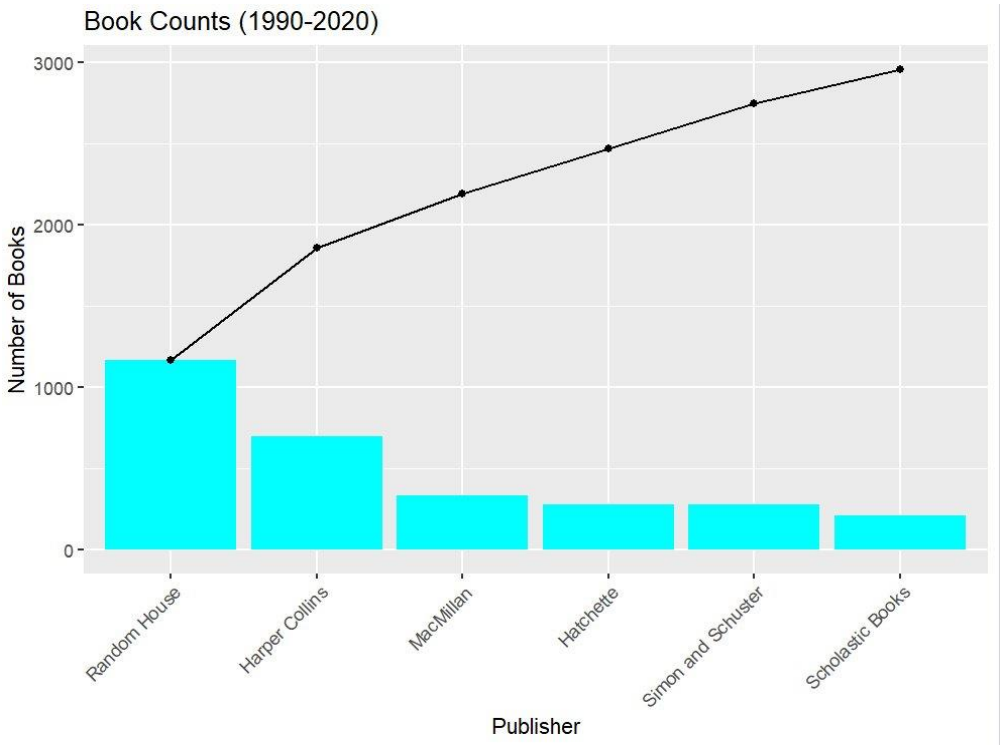

Figure 4. Pareto Chart of Book Counts by Publisher (125+ books onl

Figure 5: Top 10 Publishers by Number of Books

The bar chart of the top 10 publishers provides a broader view, including all publishers regardless of the 125-book threshold used in the Pareto chart. Bars are ordered from largest to smallest for easy visual comparison.

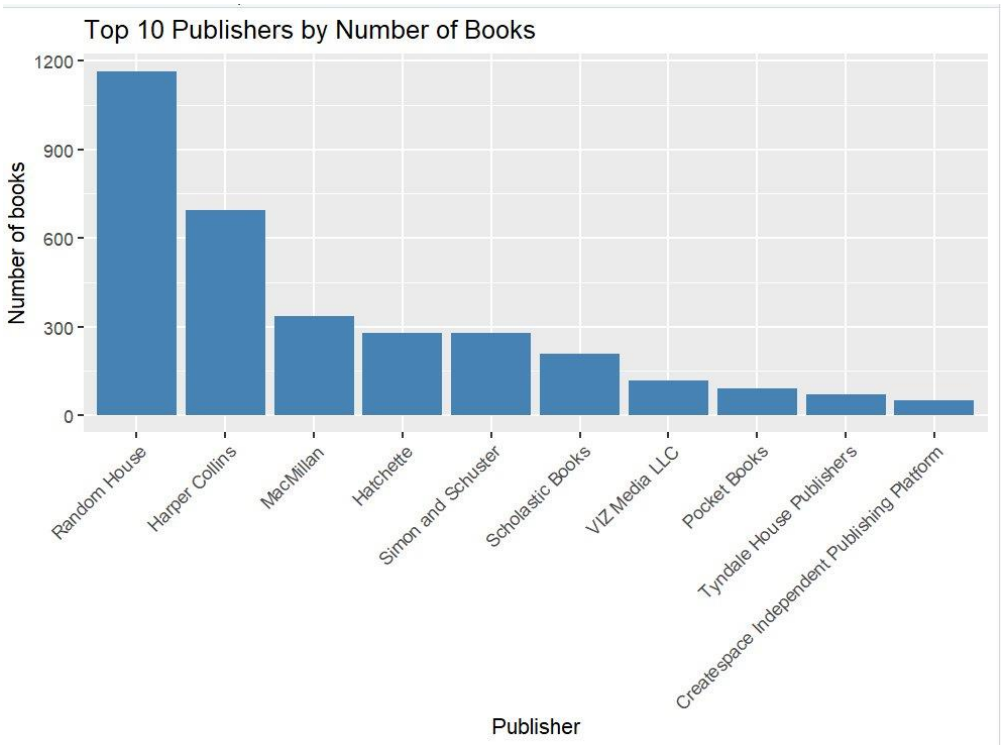

Figure 5. Top 10 Publishers by Number of Books

As shown in Figure 5, Random House leads with approximately 1,175 books, followed by Harper Collins with approximately 700 books. MacMillan, Hatchette, Simon and Schuster each contributed between 280 and 320 books. The remaining publishers are Scholastic Books, VIZ Media LLC, Pocket Books, Tyndale House Publishers, and Createspace Independent Publishing Platform contribute fewer than 200 books each. The steep drop-off after the top two publishers reinforces the concentration observed in the Pareto chart.

Figure 6: Average Book Rating Per Year (1990-2020)

This line plot tracks how the mean book rating changed year by year. Unlike Figure 3 which counted books, this chart computes the arithmetic mean of all ratings for books first published each year , making it a measure of central tendency over time.

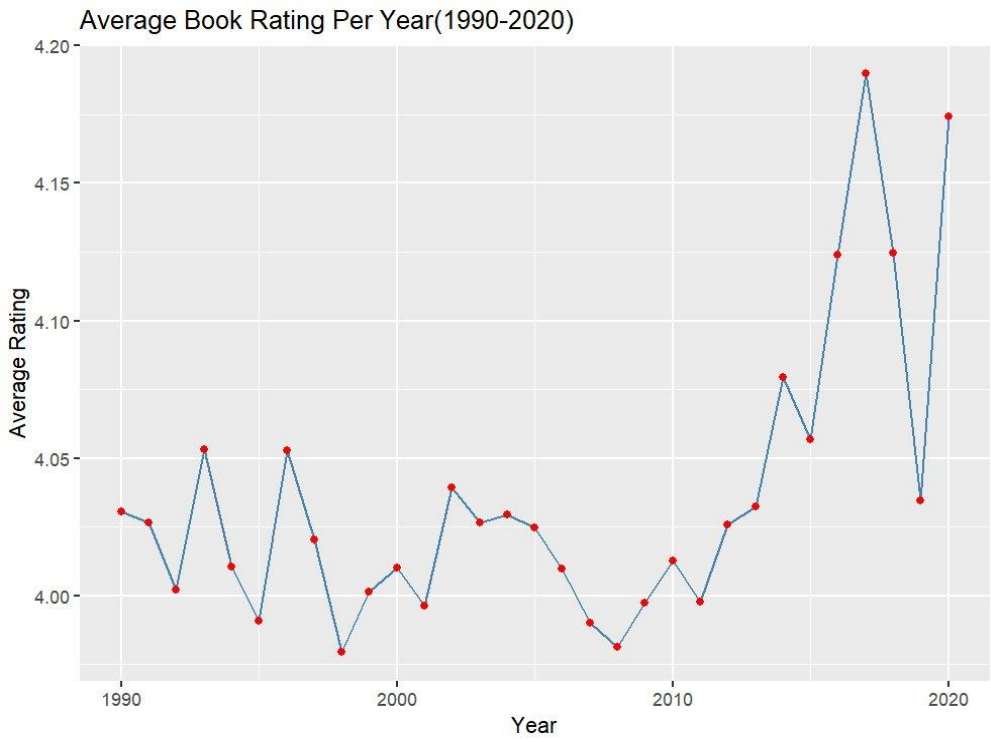

Figure 6. Average Book Rating Per Year (1990-2020)

As shown in Figure 6, average ratings fluctuated between approximately 3.97 and 4.19 across the full period. From 1990 to approximately 2014, ratings hovered near 4.00-4.05 with moderate variability. A notable upward shift appears from around 2015 onward, peaking at approximately 4.19 in 2017. This recent increase likely reflects a positive selection bias, only the most popular newer books have accumulated enough ratings to be included, artificially inflating the average for recent years. Overall, the consistently high ratings confirm that Goodreads maintains a favorable rating culture throughout the entire period.

Conclusion and Recommendations

The analysis of the Goodreads dataset yielded several meaningful insights into reading trends and publishing patterns from 1990 to 2020. The data demonstrates that Goodreads users consistently rate books positively, with a mean rating of 4.02 out of 5.00 and an average liked percentage of 92.87%. This reflects a platform culture in which readers tend to engage with books they are already predisposed to enjoy, resulting in favorably skewed ratings across the board.

The publication trend analysis revealed peak book activity around 2010-2012, corresponding to the rapid growth of Goodreads as a platform. The sharp decline in book counts after 2012 is a data artifact rather than a true publishing trend. Newer books have simply had less time to accumulate the volume of ratings needed to appear prominently in the dataset. Publishers and authors should therefore focus on building early Goodreads engagement at the time of a book's release to maximize long-term visibility.

The publisher analysis confirmed significant concentration in the industry, with Random House and Harper Collins dominating the dataset by a wide margin. Future analysis could explore the relationship between number of ratings and average ratings or examine patterns by genre or language to uncover deeper reader preference trends across global audiences.

Works Cited

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